Curriculum Vitae

Miguel A Mendez

Personal Details



Family name:	Mendez
First name:	Miguel Alfonso
Date and place of birth:	31/10/1987, Maracay (Venezuela)
Nationality:	Italian
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Current Position

- **Honorary Lecturer (interim appointment)**, Service d'Automatique et d'Analyse des Systèmes (SAAS), Brussels School of Engineering, Université libre de Bruxelles (ULB), Brussels, Belgium (February 2024–present).
- **Associate Professor**, von Karman Institute for Fluid Dynamics (VKI), Environmental & Applied Fluid Dynamics Department, Sint-Genesius-Rode, Belgium (May 2023–present).
- **Profesor Honorifico**, Experimental Aerodynamics and Propulsion Lab, Universidad Carlos III de Madrid, Leganés, Spain (February 2025–present).

Education

- **PhD in Engineering**, Université libre de Bruxelles (2013–2018). Thesis: *Dynamics of Gas Jet Impinging on Falling Liquid Films*. Supervisor: Jean Marie Buchlin (VKI).
- **Research Master in Experimental Fluid Mechanics**, von Karman Institute (2012–2013). Graduated with honours; von Karman Prize.
- **MSc in Energy and Nuclear Engineering**, University of Florence (2009–2012). GPA 110/110 *cum laude*.
- **Erasmus, 5th Year in Industrial Engineering**, ETSI Sevilla (2010–2011). GPA 8.9/10.
- **BSc in Mechanical Engineering**, University of L'Aquila (2006–2009). GPA 107/110.

Previous Positions

- **Assistant Professor**, VKI, Environmental & Applied Fluid Dynamics Department (2019–2023).
- **Postdoctoral Researcher**, VKI, Environmental & Applied Fluid Dynamics Department, and Université Paris 13, France (2018–2019).

Awards & Distinctions

Emerging Leader 2022, by *Measurement Science and Technology*. **Outstanding Paper 2020**, by *Measurement Science and Technology*. **Solvay PhD Thesis Award** for Research Excellence (2019). **AFVL PhD Thesis Award** for laser diagnostics in fluid mechanics, by AFVL. **Best Oral Presentation at the Annual VKI PhD Symposium 2015-2016. Best Oral Presentation at the 10th**

Experimental Fluid Mechanics Conference 2015 Theodore von Karman Prize (best performance at the VKI Research Master program)

Research Output and Funding

- **Research Output:** 69 published journal papers (plus 4 under review), 110 conference contributions, 10 book chapters and 2 edited books (full list at the end of this CV).
- **Impact** (Google Scholar, October 2026): 2,019 citations (1,798 since 2021); h-index 25; i10-index 44.
- **Funding (last 5 years):** ∼3.1 M€ as PI (including 1.5 M€ for an ERC Starting Grant) and ∼3.4 M€ as collaborator on specific project work packages.

Teaching (A.Y. 2025/2026)

(with starting year of each teaching activity)

- Fundamentals of Fluid Dynamics (22 h, 2.5 ECTS), VKI. *Started in 2017 as DEFM, adapted in 2019*
- Signal Processing, Module II (12 h, 1.5 ECTS), VKI. *Started in 2019*
- Introduction to Measurement Techniques (25% of the course), VKI. *Started in 2017*
- Tools for Scientific Computing (12 h, 1 ECTS), VKI *Started in 2022*
- Data-Driven Modal Analysis (12 h, 1 ECTS), VKI *Started in 2022*
- Machine Learning for Fluid Dynamics, Module I (12 h, 1 ECTS) at VKI. *Started as Data Driven Fluid Mechanics (DDFM) in 2021, adapted in 2025.*
- Machine Learning for Fluid Dynamics, Module II (24 h, 2 ECTS), VKI. *Started as DDFM since 2021, adapted in 2025.*
- Digital Twinning, Data Assimilation and Control (18 h, 2 ECTS), VKI. *Started in 2024.*
- Hands-on Machine Learning for Fluid Dynamics (32 h, 3 ECTS), VKI. *Started in 2021.*
- *Data-Driven Engineering (MECAH419)* (10 h), ULB. *Started in 2022*

Service to the Profession and Society

- Reviewer for: *Data-Centric Engineering*, *Journal of Fluid Mechanics*, *Physics of Fluids*, *Experimental Thermal and Fluid Science*, *Review of Scientific Instruments*, *Chemical Engineering Science*, *International Journal of Multiphase Flows*, *Industrial & Engineering Chemistry Research*, *Journal of Coatings Technology and Research*, *European Journal of Operational Research*.
- Organizer/director and main lecturer: *Hands-on Machine Learning for Fluid Dynamics* (yearly since 2021, >700 participants so far).
- Co-organizer/director: VKI Lecture Series on PIV/LPT (2021, 2024); *Introduction to Measurement Techniques* (2020); *Data Driven Fluid Mechanics* LS (2020, >230 participants; lecture notes published by Cambridge University Press); *Machine Learning for Fluid Dynamics* LS (2024, >300 participants; lecture notes published by von Karman Institute); *Flow Control* LS (2026).
- Scientific Committee member: *European Coating Symposium*, *AI Fluids*, *International Symposium on Flow Visualization (ISFV)*.

Supervision and PhD Committee Service

- I currently supervise 11 PhD students (8 as PI and 3 in international co-supervision), with seven defenses completed in 2023–2025. Details in Table 1.
- Visiting PhDs (2023–2026): L. Kyriakidis (DLR), T. Werner (TU Darmstadt), D. Gligor (UPM), I. Tirelli (UC3M), E. Saccaggi (PoliTo), J. Wang (Monash Univ.), D. Grasev (UoD), M. Belda (Czech Tech. Univ.), Z. Hoffman (RMIT, Australia), A. Pinardi (PoliMi), T. Giammaidoni (Univ. Perugia).
- I currently supervise 4 Research Master students at VKI (26 supervised in the last 5 years). 100+ MSc theses/internships supervised.
- Member of 11 PhD Committees: Kamila Zdybal (ULB), Mushin Akkurt (VKI-UGhent), Sagar Adatreo (TU Delft), Antonio Colanera (UniNapoli), Hang Wang (ULB), Michele Quattromini (Univ. Paris-Saclay), Ivan Kharsansky (ENSTA Paris), Enrico Amico (PoliTo), Riccardo Margheriti (PoliMi), Lorenzo Giunti (UniFI), Adrian Grille Guerra (TU Delft).

Outreach and Engagement

- **Public outreach:** Contributor to the VKI Open Days, presenting fluid-dynamics demonstrations and research activities to the general public.
- **Science communication:** Featured in a popular podcast on 'Engineering & PhD Life,' discussing fluid-dynamics research, PhD training, and career perspectives (link), and in a recorded panel organized by NCP-FNRS to share feedback and strategic advice for prospective European-grant applicants (ERC, Horizon Europe) (link).
- **Open-source software and training:** Developer of the open-source tools MODULO (MODal mUltiscale pOd, see github) for modal decomposition and data-driven reduced-order modelling and SPICY (meshlesS Pressure from Image veloCimetrY, see github) for physics-constrained radial-basis-function regression for scattered data. Both tools are accompanied by public tutorials and instructional videos for the research community (see MODULO on youtube, see SPICY on youtube).

International and National Scientific Links

Besides my strong ties with UCM3 (**Universidad Carlos III de Madrid**) through my joint position and close interactions with S. Discetti and A. Ianiro, I maintain active collaborations with several international groups via student mobility, seminars, PhD committees, and joint proposals. These include close links with: A. Gosset and M. Lema (**Universidade da Coruña, UDC**); G. Cafiero, J. Serpieri, F. Avallone, G. Di Cicca and S. Pieraccini (**Politecnico di Torino**); A. Sciacchitano, F. Scarano, A. Koa, and C. Falsetti (**Delft University of Technology, TU Delft**); M. Manolesos and K. Giannakoglou (**National Technical University of Athens, NTUA**); F. Sanfedino and A. F. Urbano (**ISAE–SUPAERO**); P. Salgado, L. González, and S. Le Clainche (**Universidad Politécnica de Madrid, UPM**); Josip Basic (**University of Split**); O. Semeraro (**CNRS–LISN**); J. Hnidka (**University of Defence**); S. Brunton (**University of Washington**); B. Noack (**Shenzhen University**), T. Pagliaroli (**UNICUSANO**), R. Camussi (**UNIROMA3**), L. Magri (**Imperial College London**); L. Pastur (**ENSTA Paris**); P. Cinnella (**Sorbonne University**); M. Bähr, S. Görtz, and P. Bekemeyer (**German Aerospace Center, DLR**); B. McKeon (**Stanford University**); P. Jordan (**Institut P', CNRS–Université de Poitiers**); O. San (**University of Tennessee**); and B. Dias (**NASA Ames Research Center**), among others. In Belgium, my closest links are with E. Garone, B. Scheid and A. Parente (**ULB**); Y. Bartosiewicz (**UCLouvain**); J. Degroote (**Ghent University**); J. Helsen (**VUB**); D. Seveno and R. Vetrano (**KU Leuven**); and J. Steelant (**KU Leuven / ESA**), complemented by broader interactions with other colleagues nationwide. I aim to further expand this network through my position at ULB.

Name	University	Co-supervisor	Funding	Year	2023	2024	2025	2026	2027	2028
Domenico Fiorini	KULeuven	D. Seveno	FWO/ESA	Defended						
Fabio Pino	ULB	B. Scheid	FRIA/AM	Defended						
Matilde Fiore	UCLouvain	Y. Bartosiewicz	FRIA/SCK	Defended						
David Barreiro	UCoruna	A. Gosset	UCoruna/AM	Defended						
Romain Poletti	UGhent	J. Degroote	FWO	Defended						
Pedro Marques	ULB	B. Scheid	FRIA/ESA	Defended						
Jan van Den Berghe	UCLouvain	Y. Bartosiewicz	FRIA	Defended						
Lorenzo Schena	VUB	J. Helsen	FWO	5th						
Manuel Ratz	ULB	A. Parente	FRIA/ERC	4th						
Damien Rigutto	ULB	B. Scheid	FRIA/AM	4th						
Umberto Zucchelli	ISAE	F. Sanfedino	ESA	4th						
Francisco Monteiro	UC3M	–	VLAIO/ERC	3rd						
Samuel Ahizi	UC3M	–	VLAIO/ERC	3rd						
Sebastiano Randino	ULB	E. Garone	FRIA/ERC	3rd						
David Grasev	UoD	J. Hnidka	ERC/UoD	3rd						
Tommaso De Maria	ULB	E. Garone	ERC	2nd						
Yannick Lecomte	ULB	E. Garone	ERC	2nd						

Table 1: Summary of PhD students and project timeline (gray = active years; light blue = current academic year). The team also includes three postdoctoral researchers: A. Grava, R. Poletti and J. Van den Berghe.

Selected Invited Lectures, Seminars and Keynotes (2022–2026)

1. **Seminar at the techMathGroup, Institute of Thermomechanics of the Czech Academy of Sciences: "Research and Collaboration Opportunities at the von Karman Institute and the Machine Learning for Fluid Systems Group"**, hosted by Dr. M. Isoz, LinkedIn post, Prague, September 2026.
2. **PhD Course at UC3M Doctoral School (Aerospace Engineering)** "From Optimal Control to Learning-Based Control: A Unified Perspective on Adaptive Control and Reinforcement Learning", Leganés (Madrid), June 2026, link.
3. **Seminar at the MLG group, ULB: "Reinforcement Twinning: From Digital Twins to Acting Models in Thermo-Fluid Engineering"**, invited by Prof. G. Bontempi, announcement, Brussels, March 2026.
4. **Lectures at the EA2 "Advanced Post Processing Techniques Summer School": AI in Fluid Dynamics — Meshless Modal Analysis of Scattered Data**, invited by Prof. Avallone, event page, Torino, July 2025.
5. **Lectures at UC3M Doctoral School** "An Introduction to Statistical and Variational Methods for Assimilation and Real Time Modeling", invited by Prof. S. Discetti, Madrid, Jun 2025, link.
6. **Keynote Lecture on Reinforcement Twinning at the 1st AI and Fluid Mechanics Conference**, invited by Prof. Gavaises, programme, Chania, May 2025.
7. **Seminar on Machine Learning for Fluid Dynamics at BrIAS**, invited by Prof. E. Garone, video, Brussels, February 2025.
8. **Seminar on Reinforcement Twinning at the WP3 Cypher Meeting**, invited by Prof. A. Parente, event page, Madrid, February 2025.
9. **Seminar on Meshless and Binless Super-resolution for PTV**, Exp in Fluids Seminar, invited by Prof. Tropea, video, Online, November 2024.
10. **Seminar on Engineering AI at the workshop "Bridging Knowledge: Artificial Intelligence"**, organized by ASI, invited by Dr. A. Turchi, agenda, Rome, May 2024.
11. **Seminar on "Trends and Challenges in Scientific Machine Learning for Engineering"**, invited by Dr. Esposito, ASI Online Seminar Series, event page, March 2024.
12. **Keynote Lecture on Reinforcement Twinning at the 1st Cypher General Meeting**, invited by Prof. A. Parente, meeting page, Ljubljana, April 2024.
13. **Keynote Lecture on Machine Learning for Fluid Dynamics at the 12th AIAC**, invited by Prof. Özgen, METU, Ankara, September 2023, conference site.
14. **Lecture on Machine Learning for Fluids at the Short Course on CFD of ICCS**, invited by Dr. Shane, London, September 2023, course programme.
15. **Three Lectures on Deep Learning for Turbulence Modeling at the FJOH Summer School**, invited by Prof. R. Stieglitz, Karlsruhe, August 2023, school page, video.
16. **Seminar at Extrality: "Trends and Challenges in Scientific Machine Learning for Fluid Dynamics"**, invited by Dr. Mazari, Online, September 2023, event link.
17. **Seminar on Machine Learning for Engineering at IMT Nord Europe**, invited by Prof. Lacassagne, Douai, June 2023.
18. **Five Lectures on Data-Driven Modal Analysis at CAMS–AUB**, invited by Prof. Najem, Beirut, February 2023, programme, video.
19. **Lecture at JT51 on Data Processing**, invited by Prof. Zimmer, Paris, November 2022, event page.

20. **Seminar at City, University of London: "Data-Driven Fluid Mechanics"**, invited by Prof. Marinos Manoles, London, October 2022, event page.

21. **Seminar at KTH: "Comparative Analysis of Machine Learning Methods for Active Flow Control"**, invited by Prof. R. Vinuesa, Stockholm, September 2022, event page.

22. **Seminar at Universidade da Coruña (UDC) on Machine Learning for Fluid Dynamics**, invited by Prof. Gosset, Ferrol, September 2022, event page.

23. **Keynote Lecture on Data-Driven Methods for Image Velocimetry at the CFTL 2022 Meeting**, invited by Prof. Vetrano, Leuven, September 2022, meeting page.

24. **Lecture on Model-Free Control Methods at the Workshop on Complex Fluids, Centro Enrico Fermi**, invited by Prof. Biferale, Rome, July 2022, programme.

List of Publications

Under Review

[1] M. Belda, L. Schena, R. Poletti, M. Isoz, T. Hyhlik, and M. A. Mendez. "A Fast Spectral Formulation of the Multiscale Proper Orthogonal Decomposition". Submitted to *Computer Physics Communications*. 2026. arXiv: 2604.12077.

[2] R. Poletti, L. Schena, L. Koloszar, J. Degroote, and M. A. Mendez. "Reinforcement Twinning for hybrid control of flapping-wing drones". Submitted to *Data-Centric Engineering*. 2025. arXiv: 2505.18201.

[3] P. A. Marques, S. Ahizi, and M. A. Mendez. "Physics-integrated neural network modeling of heat and mass transfer in cryogenic liquid storage under static and dynamic conditions". Submitted to *Energy*. 2026. arXiv: 2609.00019.

[4] U. Zucchelli, M. A. Mendez, A. Urbano, S. Vincent-Bonnieu, P. Wenderski, and F. Sanfedino. "A partitioned fluid-structure interaction solver for two-phase sloshing and flexible spacecraft dynamics". Submitted to *Aerospace Science and Technology*. 2026. arXiv: 2609.28355.

Journal Articles Published

[5] R. Poletti, E. Bombardi, L. Koloszar, M. A. Mendez, and J. Degroote. "Characterisation and Extension of a Rigid Body Dynamics Solver Coupled with OpenFOAM® for Flight Performance Analysis of Flapping-wing Drones". In: *OpenFOAM® Journal* 6 (Oct. 2026), pp. 159–186. DOI: 10.51560/ofj.v6.150. arXiv: 2510.24518.

[6] F. Monteiro, T. De Maria, S. Ahizi, R. Abarca, G. C. A. Caridi, and M. A. Mendez. "Regime maps for sloshing in horizontal cylindrical tanks under vertical acceleration". In: *Physical Review Fluids* (2026). DOI: 10.1103/t48x-53br. arXiv: 2512.02540.

[7] J. Van den Berghe, M. A. Mendez, and Y. Bartosiewicz. "On the choking mechanism in supersonic ejectors: a one-dimensional analysis of Reynolds-averaged Navier–Stokes simulations". In: *Journal of Fluid Mechanics* 1041 (Aug. 2026), A8. DOI: 10.1017/jfm.2026.11894. arXiv: 2510.23385.

[8] P. A. Marques, A. Simonini, B. Scheid, and M. A. Mendez. "Sloshing-induced pressure drop in cryogenic propellant tanks: an experimental analysis with flow visualization". In: *International Journal of Heat and Mass Transfer* 271 (Dec. 2026), p. 129320. DOI: 10.1016/j.ijheatmasstransfer.2026.129320.

[9] M. Fiore, A. Bresciani, M. A. Mendez, and J. van Beeck. "Predicting wind loads on container ships in harbour environments through multi-fidelity modelling". In: *Ocean Engineering* 364 (Aug. 2026), p. 126928. DOI: 10.1016/j.oceaneng.2026.126928. arXiv: 2604.22882.

[10] D. Rigutto, M. Ratz, and M. A. Mendez. "A meshless data-tailored approach to compute statistics from scattered data with adaptive radial basis functions". In: *Experiments in Fluids* 67 (Aug. 2026), p. 103. DOI: 10.1007/s00348-026-04257-4. arXiv: 2511.20449.

[11] D. Barreiro-Villaverde, A. Cantiani, and M. A. Mendez. "On the complex interplay of temperature, phase change and natural convection in self-pressurization—an investigation using segregated modeling". In: *International Journal of Heat and Mass Transfer* 269 (Nov. 2026), p. 129150. DOI: 10.1016/j.ijheatmasstransfer.2026.129150. arXiv: 2512.13349.

[12] M. Ratz, A. Parente, and M. A. Mendez. "Meshless data-driven decompositions with RBF-based inner products". In: *Computers and Fluids* 315 (July 2026), p. 107133. DOI: 10.1016/j.compfluid.2026.107133. arXiv: 2511.03264.

[13] S. Randino, L. Schena, N. Coudou, E. Garone, and M. A. Mendez. "Nonlinear system identification for model-based control of waked wind turbines". In: *Data-Centric Engineering* 7 (June 2026). DOI: 10.1017/dce.2026.10051. arXiv: 2510.07336.

[14] A. Parente, G. Sorrentino, P. Sabia, C. Mounaïm-Rousselle, V. Jozsa, M. de Joannon, F. Battin Leclerc, A. Attili, F. Ferraro, P. E. Lapenna, D. Mira, A. Cuoci, G. Magnotti, A. K. Doan, K. Bizon, P. Cinnella, M. A. Mendez, L. Vervisch, E. Petlenkov, A. Onile, P. Gorm Larsen, R. Ragucci, A. Kazagic, A. Della Rocca, J. Leicher, I. El Helou, T. Lima Silva, G. Skevis, and C. Caccamo. "A roadmap for decarbonizing energy-intensive industries with renewable synthetic fuels and digital technologies". In: *Journal of Physics: Energy* (2026). DOI: 10.1088/2515-7655/ae67a4.

[15] F. Pino, E. Fracchia, B. Scheid, and M. A. Mendez. "Integral modelling and reinforcement-learning control of 3D liquid metal coating on a moving substrate". In: *Physical Review Fluids* 11 (2026), p. 044003. DOI: 10.1103/77zf-ks57. arXiv: 2503.14270.

[16] S. Ahizi, F. Monteiro, R. Abarca, and M. A. Mendez. "Real-time identification of parametric sloshing-induced heat and mass transfer in a horizontally oriented cylindrical tank". In: *International Journal of Heat and Mass Transfer* 264 (2026), p. 128707. DOI: 10.1016/j.ijheatmasstransfer.2026.128707. arXiv: 2510.19540.

[17] I. Tirelli, M. A. Mendez, A. Ianiro, and S. Discetti. "Meshless super-resolution of scattered data via constrained radial basis functions and K-nearest-neighbors-driven densification". In: *Physical Review Fluids* 11.2 (Feb. 2026), p. 024902. DOI: 10.1103/nvlc-1cmn. arXiv: 2503.04630.

[18] L. Schena, W. Munters, J. Helsen, and M. A. Mendez. "POD-Based Sparse Stochastic Estimation of Dynamic Wind Turbine Blade Deflections". In: *Journal of Sound and Vibration* (Mar. 2026), p. 119738. DOI: 10.1016/j.jsv.2026.119738. arXiv: 2504.08505.

[19] P. A. Marques, S. Ahizi, F. Monteiro, B. Scheid, and M. A. Mendez. "On the scaling of heat and mass transfer in cryogenic propellant tanks: A model-based and experimental analysis under static conditions and lateral sloshing". In: *Applied Thermal Engineering* 279 (2025), p. 127751. DOI: 10.1016/j.applthermaleng.2025.127751.

[20] L. Kyriakidis, M. Bähr, and M. A. Mendez. "Enhanced hybrid algorithm based on Bayesian optimization and Interior Point OPTimizer for constrained optimization". In: *Optimization and Engineering* (2025). DOI: 10.1007/s11081-025-09975-y.

[21] I. Tirelli, M. A. Mendez, A. Ianiro, and S. Discetti. "A meshless method to compute the proper orthogonal decomposition and its variants from scattered data". In: *Proceedings of the Royal Society A* (2025). DOI: 10.1098/rspa.2024.0526. arXiv: 2407.03173.

[22] M. Fiore, E. Saccaggi, L. Koloszar, Y. Bartosiewicz, and M. A. Mendez. "Data-driven turbulent heat flux modeling with inputs of multiple fidelity". In: *Physical Review Fluids* 10.3 (2025), p. 034606. DOI: 10.1103/PhysRevFluids.10.034606. arXiv: 2409.03395.

[23] F. Monteiro, P. Marques, A. Simonini, L. Carbonelle, and M. A. Mendez. "Experimental characterization of non-isothermal sloshing in microgravity". In: *Experimental Thermal and Fluid Science* 166 (2025), p. 111476. DOI: 10.1016/j.expthermflusci.2025.111473. arXiv: 2410.06590.

[24] F. Pino, M. A. Mendez, and B. Scheid. "Pareto front of the magnetic wiping process in dip coating". In: *Journal of Engineering Mathematics* (2025). DOI: 10.1007/s10665-025-10426-x. arXiv: 2406.14110.

[25] J. Van den Berghe, M. A. Mendez, and Y. Bartosiewicz. "An extension of the compound flow theory with friction between the streams and at the wall". In: *Journal of Fluid Mechanics* 1000 (2024), A85. DOI: 10.1017/jfm.2024.1041. arXiv: 2401.07236.

[26] F. Pino, M. A. Mendez, and B. Scheid. "Linear stability analysis of a vertical liquid film over a moving substrate". In: *Journal of Fluid Mechanics* 1000 (2024), A57. DOI: 10.1017/jfm.2024.940.

[27] D. Barreiro-Villaverde, A. Gosset, and M. A. Mendez. "On the coupling instability of a gas jet impinging on a liquid film". In: *Journal of Fluid Mechanics* 992 (2024), A11. DOI: 10.1017/jfm.2024.553. arXiv: 2309.15502.

[28] F. Pino, M. A. Mendez, and B. Scheid. "Absolute and convective instabilities in a liquid film over a substrate moving against gravity". In: *Physical Review Fluids* 9 (2024), p. 104002. DOI: 10.1103/PhysRevFluids.9.104002. arXiv: 2312.14613.

[29] L. Kyriakidis, M. A. Mendez, and M. Bahr. "A hybrid algorithm based on Bayesian Optimization and Interior Point OPTimizer for optimal operation of energy conversion systems". In: *Energy* 312 (2024), p. 133416. DOI: 10.1016/j.energy.2024.133416.

[30] D. Gligor, P. A. Marques, P. S. Sanchez, J. Porter, M. A. Mendez, and J. M. Ezquerro. "Experiments on sloshing mitigation using tuned oscillating baffles". In: *Physics of Fluids* 36 (2024), p. 092122. DOI: 10.1063/5.0225917.

[31] D. Fiorini, A. Simonini, J. Steelant, D. Seveno, and M. A. Mendez. "An experimental characterization of capillary driven flows in microgravity". In: *Microgravity Science and Technology* 36 (2024), p. 61. DOI: 10.1007/s12217-024-10142-8.

[32] R. Poletti, L. Schena, D. Ninni, and M. A. Mendez. "MODULO: a python toolbox for data-driven modal decomposition". In: *Journal of Open Source Software* 9.102 (2024), p. 6753. DOI: 10.21105/joss.06753.

[33] R. Poletti, A. Calado, L. Koloszar, J. Degroote, and M. A. Mendez. "On the unsteady aerodynamics of flapping wings under dynamic hovering kinematics". In: *Physics of Fluids* 36 (2024), p. 081901. DOI: 10.1063/5.0215531. arXiv: 2408.03222.

[34] M. Ratz and M. A. Mendez. "A meshless and binless approach to compute statistics in 3D ensemble PTV". In: *Experiments in Fluids* 65 (2024), p. 142. DOI: 10.1007/s00348-024-03878-x. arXiv: 2403.11828.

[35] L. Schena, P. Marques, R. Poletti, S. Ahizi, J. Van den Berghe, and M. A. Mendez. "Reinforcement Twinning: from digital twins to model-based reinforcement learning". In: *Journal of Computational Science* 82 (2024), p. 102421. DOI: 10.1016/j.jocs.2024.102421. arXiv: 2311.03628.

[36] P. Marques, S. Ahizi, and M. A. Mendez. "Real-time data assimilation for the thermodynamic modelling of cryogenic storage tanks". In: *Energy* 302 (2024), p. 131739. DOI: 10.1016/j.energy.2024.131739. arXiv: 2310.11399.

[37] D. Gligor, C. Peromingo, P. Salgado Sanchez, J. Porter, and M. A. Mendez. "Sloshing mitigation in microgravity with moving baffles". In: *Acta Astronautica* 219 (2024), pp. 639–652. DOI: 10.1016/j.actaastro.2024.03.047.

[38] P. Sperotto, M. Ratz, and M. A. Mendez. "SPICY: a Python toolbox for meshless assimilation from image velocimetry using radial basis functions". In: *Journal of Open Source Software* 9.93 (2024), p. 5749. DOI: 10.21105/joss.05749.

[39] D. Fiorini, A. Simonini, J. Steelant, D. Seveno, and M. A. Mendez. "Dynamic wetting experiments with nitrogen in a quasi-capillary tube". In: *Physical Review Fluids* 8 (2023), p. 124004. DOI: 10.1103/PhysRevFluids.8.124004. arXiv: 2310.05490.

[40] D. Barreiro-Villaverde, A. Gosset, and M. A. Mendez. "Damping of three-dimensional waves on coating films dragged by moving substrates". In: *Physics of Fluids* 35 (2023), p. 072110. DOI: 10.1063/5.0154144. arXiv: 2305.16139.

[41] T. Ivanova, F. Pino, B. Scheid, and M. A. Mendez. "Evolution of waves in liquid films on moving substrates". In: *Physics of Fluids* 35 (2023), p. 013609. DOI: 10.1063/5.0132222. arXiv: 2203.08201.

[42] D. Fiorini, L. Carbonelle, A. Simonini, J. Steelant, D. Seveno, and M. A. Mendez. "Characterization of a capillary-driven flow in microgravity by means of optical techniques". In: *Multiphase Science and Technology* 35.3 (2023), pp. 55–66. DOI: 10.1615/MultScienTechn.2023047919.

[43] P. Marques, A. Simonini, L. Peveroni, and M. A. Mendez. "Experimental analysis of heat and mass transfer in non-isothermal sloshing using a model-based inverse method". In: *Applied Thermal Engineering* (2023), p. 120871. DOI: 10.1016/j.applthermaleng.2023.120871. arXiv: 2212.12246.

[44] A. Calado, R. Poletti, L. K. Koloszar, and M. A. Mendez. "A robust data-driven model for flapping aerodynamics under different hovering kinematics". In: *Physics of Fluids* 35 (2023), p. 047122. DOI: 10.1063/5.0139020. arXiv: 2212.09464.

[45] F. Pino, L. Schena, J. Rabault, and M. A. Mendez. "Comparative analysis of machine learning methods for active flow control". In: *Journal of Fluid Mechanics* 958 (2023), A39. DOI: 10.1017/jfm.2023.76. arXiv: 2202.11664.

[46] L. Gkimisis, B. R. Barros Dias, J. B. Scoggins, T. Magin, M. A. Mendez, and A. Turchi. "Data-driven modeling of stagnation-line flow with heat and mass transfer in hypersonic reentry". In: *AIAA Journal* 61.8 (2023), pp. 3269–3286. DOI: 10.2514/1.J062332. arXiv: 2208.06240.

[47] F. J. Mendez, M. A. Mendez, N. Sciarra, and A. Pasculli. "Multiobjective analysis of the Sand Hypoplasticity model calibration". In: *Acta Geotechnica* (2024). DOI: 10.1007/s11440-023-02191-x. arXiv: 2303.07234.

[48] J. Van den Berghe, B. R. Barros Dias, Y. Bartosiewicz, and M. A. Mendez. "A 1D model for the unsteady gas dynamics of ejectors". In: *Energy* (2022). DOI: 10.1016/j.energy.2022.126551. arXiv: 2208.07687.

[49] M. A. Mendez. "Linear and nonlinear dimensionality reduction from fluid mechanics to machine learning". In: *Measurement Science and Technology* 34.4 (2023). DOI: 10.1088/1361-6501/acaffe. arXiv: 2208.07746.

[50] A. Procaci, M. M. Kamal, M. A. Mendez, S. Hochgreb, A. Coussement, and A. Parente. "Multi-scale proper orthogonal decomposition analysis of instabilities in swirled and stratified flames". In: *Physics of Fluids* 32 (2020), p. 123103. DOI: 10.1063/5.0127956.

[51] M. Fiore, L. Koloszar, M. A. Mendez, M. Duponcheel, and Y. Bartosiewicz. "Turbulent heat flux modeling in forced convection flows using artificial neural networks". In: *Nuclear Engineering and Design* 399.1 (2022), p. 112005. DOI: 10.1016/j.nucengdes.2022.112005.

[52] D. Fiorini, M. A. Mendez, A. Simonini, J. Steelant, and D. Seveno. "Effect of inertia on the dynamic contact angle in oscillating menisci". In: *Physics of Fluids* 32 (2022), p. 102116. DOI: 10.1063/5.0119863. arXiv: 2208.05329.

[53] M. Ratz, D. Fiorini, A. Simonini, C. Cierpka, and M. A. Mendez. "Analysis of an unsteady quasi-capillary channel flow with time-resolved PIV and RBF-based super-resolution". In: *Journal of Coatings Technology and Research* (2022). DOI: 10.1007/s11998-022-00664-4. arXiv: 2202.13724.

[54] P. Sperotto, S. Pieraccini, and M. A. Mendez. "A meshless method to measure pressure fields from image velocimetry". In: *Measurement Science and Technology* 33 (2022), p. 094005. DOI: 10.1088/1361-6501/ac70a9. arXiv: 2112.12752.

[55] J. Dominique, J. Van den Berghe, C. Schram, and M. A. Mendez. "Artificial neural networks modeling of wall pressure spectra beneath turbulent boundary layers". In: *Physics of Fluids* 34 (2022). DOI: 10.1063/5.0083241. arXiv: 2201.03262.

[56] M. Fiore, L. Koloszar, M. A. Mendez, M. Duponcheel, and Y. Bartosiewicz. "Physics-constrained machine learning for thermal turbulence modeling at low Prandtl numbers". In: *International Journal of Heat and Mass Transfer* 194 (2022), p. 122998. DOI: 10.1016/j.ijheatmasstransfer.2022.122998. arXiv: 2201.06301.

[57] D. Barreiro-Villaverde, A. Gosset, and M. A. Mendez. "On the dynamics of the jet wiping process: numerical simulations and modal analysis". In: *Physics of Fluids* 33 (2021), p. 062114. DOI: 10.1063/5.0051451. arXiv: 2103.16116.

[58] M. Balabane, M. A. Mendez, and S. Nagem. "Koopman operator for Burgers's equation". In: *Physical Review Fluids* 6 (2021), p. 064401. DOI: 10.1103/PhysRevFluids.6.064401. arXiv: 2007.01218.

[59] F. J. Mendez, A. Pasculli, M. A. Mendez, and N. Sciarra. "Calibration of a hypoplastic model using genetic algorithms". In: *Acta Geotechnica* 16 (2021), pp. 2031–2047. DOI: 10.1007/s11440-020-01135-z.

[60] D. Ninni and M. A. Mendez. "MODULO: A software for multiscale proper orthogonal decomposition of data". In: *SoftwareX* 12 (2020), p. 100622. DOI: 10.1016/j.softx.2020.100622. arXiv: 2001.01971.

[61] M. A. Mendez, A. Gosset, B. Scheid, M. Balabane, and J.-M. Buchlin. "Dynamics of the jet wiping process via integral models". In: *Journal of Fluid Mechanics* 911 (2021), A47. DOI: 10.1017/jfm.2020.107. arXiv: 2004.13400.

[62] C. Esposito, M. A. Mendez, J. Steelant, and M. R. Vetrano. "Spectral and modal analysis of a cavitating flow through an orifice". In: *Experimental Thermal and Fluid Science* 121 (2021), p. 110251. DOI: 10.1016/j.expthermflusci.2020.110251.

[63] M. A. Mendez, D. Hess, B. Watz, and J.-M. Buchlin. "Multiscale proper orthogonal decomposition (mPOD) of TR-PIV data: a case study on stationary and transient cylinder wake flows". In: *Measurement Science and Technology* 31.9 (2020), p. 094014. DOI: 10.1088/1361-6501/ab82be. arXiv: 2001.01971.

[64] D. Bian, R. Theunissen, M. A. Mendez, and Y. Wei. "Probabilistic evaluation of streamline topologies for the detection of preferential flow configurations in PIV applications". In: *Experiments in Fluids* 61.13 (2020). DOI: 10.1007/s00348-019-2852-9.

[65] M. A. Mendez, A. Gosset, and J.-M. Buchlin. "An experimental analysis of the stability of the jet wiping process: Part II—Multiscale modal analysis of the gas jet–liquid film interaction". In: *Experimental Thermal and Fluid Science* 106 (2019), pp. 48–67. DOI: 10.1016/j.expthermflusci.2019.03.004.

[66] A. Gosset, M. A. Mendez, and J.-M. Buchlin. "An experimental analysis of the stability of the jet wiping process: Part I—Characterization of the coating uniformity". In: *Experimental Thermal and Fluid Science* 103 (2019), pp. 51–65. DOI: 10.1016/j.expthermflusci.2018.12.029.

[67] M. A. Mendez, M. Balabane, and J.-M. Buchlin. "Multi-scale proper orthogonal decomposition of complex fluid flows". In: *Journal of Fluid Mechanics* 870 (2019), pp. 988–1036. DOI: 10.1017/jfm.2019.212. arXiv: 1804.09646.

[68] M. A. Mendez, A. Enache, A. Gosset, and J.-M. Buchlin. "Experimental characterization of the jet wiping process". In: *EPJ Web of Conferences* 180 (2018), p. 02064.

[69] M. A. Mendez, M. T. Scelzo, A. Enache, and J.-M. Buchlin. "Fluidic vectoring of a planar incompressible jet flow". In: *EPJ Web of Conferences* 180 (2018), p. 02065. DOI: 10.1051/epjconf/201818002065.

[70] M. A. Mendez, M. T. Scelzo, and J.-M. Buchlin. "Multiscale modal analysis of an oscillating impinging gas jet". In: *Experimental Thermal and Fluid Science* 91 (2018), pp. 256–276. DOI: 10.1016/j.expthermflusci.2017.10.032.

[71] M. A. Mendez, B. Scheid, and J.-M. Buchlin. "Low Kapitza falling liquid films". In: *Chemical Engineering Science* 170 (2017), pp. 122–138. DOI: 10.1016/j.ces.2016.12.050.

[72] M. Mendez, M. Raiola, A. Masullo, S. Discetti, A. Ianiro, R. Theunissen, and J.-M. Buchlin. "POD-based background removal for particle image velocimetry". In: *Experimental Thermal and Fluid Science* 91 (2017), pp. 181–192. DOI: 10.1016/j.expthermflusci.2016.08.021.

[73] M. Mendez, L. Nemeth, and J.-M. Buchlin. "Measurement of liquid-film thickness via light absorption and laser tomography". In: *EPJ Web of Conferences* 144 (2016), p. 02072. DOI: 10.1051/epjconf/201611402072.

Edited Books

[1] M. A. Mendez, A. Ianiro, B. R. Noack, and S. L. Brunton, eds. *Data-Driven Fluid Mechanics: Combining First Principles and Machine Learning*. Cambridge University Press, Jan. 2023. ISBN: 9781108842143. DOI: 10.1017/9781108896214.

[2] M. A. Mendez and A. Parente, eds. *Machine Learning for Fluid Dynamics*. von Karman Institute for Fluid Dynamics, Nov. 2025. ISBN: 978-2875162090.

Book Chapters

[1] M. A. Mendez, R. Poletti, L. Schena, S. Randino, F. Monteiro, T. De Maria, and Y. Lecomte. "From Optimization to Learning: a unified perspective on control algorithms". In: *Flow Control*. Ed. by M. A. Mendez and S. Discetti. Linked to the VKI course *Digital Twinning, Data Assimilation and Control*. von Karman Institute for Fluid Dynamics, 2026. URL: https://www.mendezma.com/assets/pdf/DTAC_Part_2.pdf.

[2] M. A. Mendez, R. Poletti, L. Schena, S. Randino, F. Monteiro, T. De Maria, and Y. Lecomte. "Reinforcement Twinning: Towards a Bidirectional Learning Framework for Model Updating and Policy Optimization". In: *Flow Control*. Ed. by M. A. Mendez and S. Discetti. Linked to the VKI course *Digital Twinning, Data Assimilation and Control*. von Karman Institute for Fluid Dynamics, 2026. URL: https://www.mendezma.com/assets/pdf/DTAC_Part_3.pdf.

[3] M. A. Mendez. "Fundamentals of Regression". In: *Machine Learning for Fluid Mechanics*. Ed. by M. A. Mendez and A. Parente. von Karman Institute for Fluid Dynamics, 2025. arXiv: 2512.01920.

[4] M. A. Mendez, J. van den Berghe, M. Ratz, M. Fiore, and L. Schena. "Learning with Physics Constraints". In: *Machine Learning for Fluid Mechanics*. Ed. by M. A. Mendez and A. Parente. von Karman Institute for Fluid Dynamics, 2025. arXiv: 2512.00104.

[5] M. A. Mendez, R. Poletti, and L. Schena. "Fundamentals of Dimensionality Reduction". In: *Machine Learning for Fluid Mechanics*. Ed. by M. A. Mendez and A. Parente. von Karman Institute for Fluid Dynamics, 2025.

[6] M. A. Mendez, M. Ratz, and D. Rigutto. "Statistical Methods and Modal Decompositions for Gridded and Scattered Data: Meshless Statistics and Meshless Data-Driven Modal Analysis". In: *Particle Image Velocimetry*. Ed. by S. Discetti and M. A. Mendez. von Karman Institute for Fluid Dynamics, 2024. arXiv: 2502.04765.

[7] M. A. Mendez. "Continuous and Discrete Linear Time-Invariant Systems". In: *Data-Driven Fluid Mechanics: Combining First Principles and Machine Learning*. Ed. by M. A. Mendez, A. Ianiro, B. R. Noack, and S. L. Brunton. Cambridge University Press, 2023. DOI: 10.1017/9781108896214.003. arXiv: 2208.11891.

[8] M. A. Mendez. "Generalized and Multiscale Modal Analysis". In: *Data-Driven Fluid Mechanics: Combining First Principles and Machine Learning*. Ed. by M. A. Mendez, A. Ianiro, B. R. Noack, and S. L. Brunton. Cambridge University Press, 2023. DOI: 10.1017/9781108896214.004. arXiv: 2208.12630.

[9] M. A. Mendez. “Statistical Treatment, Fourier and Modal Decompositions”. In: *Fundamentals and Recent Advances in Particle Image Velocimetry and Lagrangian Particle Tracking*. Ed. by S. Discetti and M. A. Mendez. von Karman Institute for Fluid Dynamics, 2021. arXiv: 2201.03847.

[10] M. A. Mendez, M. Fiore, and F. Pino. “Machine Learning for Fluid Mechanics: Challenges, Opportunities and Perspectives”. In: *Optimization Methods for Computational Fluid Dynamics*. Ed. by J. Périaux and T. Verstraete. von Karman Institute for Fluid Dynamics, 2022. arXiv: 2202.12577.

Conference Proceedings

[1] G. Glabeke, L. Schena, W. Munters, J. Helsen, M. A. Mendez, T. De Mulder, and J. van Beeck. “Detecting Low-Level Jets over the Belgian North Sea from Sparse In-Situ Measurements using Temporal Convolutional Networks”. In: *Journal of Physics: Conference Series 3224(2)*. 2026, p. 022044.

[2] L. Schena, G. Glabeke, W. Munters, T. De Mulder, J. Helsen, J. van Beeck, and M. A. Mendez. “Aeroelastic Response of Offshore Wind Turbines to Low-Level Jets Using OpenFAST”. In: *Journal of Physics: Conference Series 3224(4)*. 2026, p. 042038.

[3] B. Dello Iacono, L. Schena, S. Stipa, J. Helsen, M. A. Mendez, and W. Munters. “Calibration and validation of FAST.Farm against LES for floating offshore wind farms”. In: *Journal of Physics: Conference Series 3224(3)*. 2026, p. 032130.

[4] J. Gebel, L. Schena, A. Rezaei, T. Verstraeten, P.-J. Daems, A. R. Nejad, and J. Helsen. “Impact of Curtailed Operation on the Fatigue Life of Offshore Wind Turbine Components”. In: *Journal of Physics: Conference Series 3224(6)*. 2026, p. 062050.

[5] L. Schena, W. Munters, J. Helsen, and M. A. Mendez. “Reinforcement Learning–Based Adaptive Torque Control of a Wind Turbine: An Experimental Study”. In: *Journal of Physics: Conference Series 3224(5)*. 2026, p. 052036.

[6] M. Ratz, C. Schram, A. Sciacchitano, and M. A. Mendez. “Experimental Characterization of an Accelerating and Decelerating Propeller at Low Reynolds Numbers”. In: *Vertical Flight Society’s 82nd Annual Forum & Technology Display (VFS2026)*. West Palm Beach, USA, 2026. DOI: 10.4050/F-0082-2026-0088.

[7] M. Ratz, Y. Lecomte, C. Schram, A. Sciacchitano, and M. A. Mendez. “Phase-locked PIV measurements of an accelerating propeller blade at low Reynolds numbers”. In: *22nd International Symposium on Applications of Laser and Imaging Techniques to Fluid Mechanics (LXLASER26)*. Lisbon, Portugal, 2026.

[8] K. Bukowski, T. De Maria, S. Ahizi, and M. A. Mendez. “Data-driven modelling and state estimation of sloshing dynamics from sparse measurements: an experimental and numerical study”. In: *XXVII Fluid Mechanics Conference (FMC 2026)*. Łódź, Poland, 2026.

[9] K. Bukowski, T. De Maria, S. Ahizi, and M. A. Mendez. “Data-driven modelling and state estimation of sloshing dynamics from sparse measurements: an experimental and numerical study”. In: *2nd International Symposium on AI and Fluid Mechanics (AIFLUIDs)*. Chania, Greece, 2026.

[10] D. Rigutto, M. Ratz, and M. A. Mendez. “A meshless data-tailored approach to compute statistics from scattered data with adaptive radial basis functions”. In: *22nd International Symposium on Application of Laser and Imaging Techniques to Fluid Mechanics*. Lisbon, Portugal, 2026.

[11] S. Randino, L. Schena, E. Garone, and M. A. Mendez. “A Self-Calibrating Wind Turbine Digital Twin for Rotor-Effective Wind-Speed Estimation and Aerodynamic-Map Identification”. In: *2nd International Symposium on AI and Fluid Mechanics (AI & Fluids 2026)*. Chania, Crete, Greece, 2026.

[12] Y. Lecomte, M. Ratz, C. Schram, and M. A. Mendez. "Experimental investigation of transient inflow and wake response of a low-Reynolds two-bladed propeller". In: *22nd International Symposium on Applications of Laser and Imaging Techniques to Fluid Mechanics (LXLASER26)*. Lisbon, Portugal, 2026.

[13] U. Zucchelli, M. A. Mendez, A. Urbano, S. V. Bonnieu, P. Wenderski, and F. Sanfedino. "Coupled Dynamics of Liquid Sloshing and Flexible Spacecraft Structures". In: *ESA International Conference on Guidance, Navigation & Control Systems and International Conference on Astrodynamics Tools and Techniques (ESA GNC & ICATT)*. Sevilla, Spain, 2026.

[14] Y. Lecomte and M. A. Mendez. "Bayesian Reinforcement Twinning: A Multi-Objective and Multi-Fidelity Framework for Reciprocal Learning Between Digital Twins and Control". In: *Data-driven active control in flows: from model-based to reinforcement learning methods*. Paris, France, 2026.

[15] Y. Lecomte, M. Ratz, L. Schena, A. Vardanyan, and M. A. Mendez. "Reinforcement twinning for attitude control of multirotor drones: an experimental proof of concept". In: *1st International Symposium on AI and Fluid Mechanics*. Chania, Greece, 2025.

[16] D. Rigutto, D. Reviglio, B. Scheid, and M. A. Mendez. "An experimental investigation of dynamic wetting at high Weber numbers: interface dynamics and flow-field analysis". In: *European Coating Symposium (ECS)*. A Coruña, Spain, 2025.

[17] D. Rigutto, B. Scheid, and M. A. Mendez. "Real-time interface reconstruction of a falling liquid film using deflectometry". In: *European Coating Symposium (ECS)*. A Coruña, Spain, 2025.

[18] M. Ratz, C. Schram, A. Sciacchitano, and M. A. Mendez. "Flow development over a low-Reynolds propeller in accelerating conditions". In: *16th International Symposium on Particle Image Velocimetry (ISPIV)*. Tokyo, Japan, 2025.

[19] M. Ratz, S. Ahizi, A. Parente, and M. A. Mendez. "A framework for meshless data-driven decompositions with RBF-based inner products". In: *2nd ERCOFTAC Workshop on Machine Learning for Fluid Dynamics*. London, United Kingdom, 2025.

[20] U. Zucchelli, M. A. Mendez, A. Urbano, S. V. Bonnieu, P. Wenderski, and F. Sanfedino. "Closed-loop control of sloshing fuel in a spinning spacecraft". In: *11th European Conference for Aeronautics and Space Sciences (EUCASS)*. Rome, Italy, 2025.

[21] U. Zucchelli, I. Valles Sánchez, and F. Sanfedino. "Modeling of Spinning Plates: Geometric Stiffening and Modal Approximation for GNC Applications". In: *28th AIDAA International Congress and 10th CEAS Aerospace Europe Conference*. Turin, Italy, 2025.

[22] T. De Maria, F. Monteiro, S. Ahizi, R. Abarca, G. C. A. Caridi, and M. A. Mendez. "Experimental analysis of sloshing-induced thermal mixing in horizontally oriented cylindrical tanks subjected to vertical acceleration". In: *11th European Conference for Aeronautics and Space Sciences (EUCASS)*. Rome, Italy, 2025.

[23] F. Monteiro, T. De Maria, S. Ahizi, R. Abarca, G. C. A. Caridi, and M. A. Mendez. "Data-driven modal decomposition and automatic classification of parametric sloshing regimes in horizontal cylindrical tanks". In: *11th European Conference for Aeronautics and Space Sciences (EUCASS)*. Rome, Italy, 2025.

[24] S. Ahizi and M. A. Mendez. "Meshless POD for sloshing-mode identification in non-canonical tank geometries". In: *AI & Fluids*. Chania, Greece, 2025.

[25] S. Randino, L. Schena, N. Coudou, E. Garone, and M. A. Mendez. "Real-time data assimilation for the digital twinning of wind farms". In: *AI & Fluids*. Chania, Greece, 2025.

[26] D. Rigutto, M. Ratz, and M. A. Mendez. "Anisotropic and multi-resolution RBFs for meshless data assimilation of scattered data". In: *AI & Fluids*. Chania, Greece, 2025.

[27] J. Van den Berghe, M. A. Mendez, and Y. Bartosiewicz. “A 1D ejector model based on the compound flow theory”. In: *2nd European Fluid Dynamics Conference (EFDC2)*. Dublin, Ireland, 2025.

[28] D. Barreiro-Villaverde, A. Gosset, M. A. Mendez, and M. Lema. “Coupling of a thin-film solver with OpenFOAM to resolve liquid–gas interactions”. In: *26th International Congress of Theoretical and Applied Mechanics (ICTAM 2024)*. Daegu, Republic of Korea, 2024.

[29] J. Van den Berghe, R. Debroeyer, M. A. Mendez, and Y. Bartosiewicz. “A one-dimensional model for supersonic ejectors based on the compound flow theory”. In: *1st Belgian Symposium of Thermodynamics*. Liège, Belgium, 2024.

[30] D. Rigutto, B. Scheid, and M. A. Mendez. “Étude expérimentale de la dynamique d’interface et du champ de vitesse autour de la ligne de contact d’une plaque immergée”. In: *18ème Congrès Francophone de Techniques Laser (CFTL)*. Lyon, France, 2024.

[31] F. Pino, B. Scheid, M. A. Mendez, and D. Papageorgiou. “Linear feedback control of a liquid film over a substrate moving against gravity”. In: *26th International Congress of Theoretical and Applied Mechanics (ICTAM 2024)*. Daegu, Republic of Korea, 2024.

[32] F. Pino, M. A. Mendez, and B. Scheid. “Absolute and convective instabilities in a liquid film over a substrate moving against gravity”. In: *European Fluid Dynamics Conference (EFDC1)*. Aachen, Germany, 2024.

[33] J. Van den Berghe, Y. Bartosiewicz, and M. A. Mendez. “Choking of parallel compressible streams with shear and wall friction”. In: *26th International Congress of Theoretical and Applied Mechanics (ICTAM 2024)*. Daegu, Republic of Korea, 2024.

[34] I. Tirelli, M. A. Mendez, A. Ianiro, and S. Discetti. “A meshless method to compute the POD and its variants from scattered data”. In: *21st International Symposium on Applications of Laser and Imaging Techniques to Fluid Mechanics (LXLASER24)*. Lisbon, Portugal, 2024.

[35] I. Tirelli, M. A. Mendez, A. Ianiro, and S. Discetti. “A full meshless algorithm for super-resolution in image velocimetry based on KNN-PTV and constrained RBFs”. In: *21st International Symposium on Applications of Laser and Imaging Techniques to Fluid Mechanics (LXLASER24)*. Lisbon, Portugal, 2024.

[36] D. Rigutto, Y. Lecomte, J.-M. Buchlin, B. Scheid, and M. A. Mendez. “Experimental investigation of the gas–liquid interface and flow-field dynamics near the contact line in an immersing flat surface”. In: *21st International Symposium on Applications of Laser and Imaging Techniques to Fluid Mechanics (LXLASER24)*. Lisbon, Portugal, 2024.

[37] M. Ratz and M. A. Mendez. “A meshless and binless approach to compute statistics in 3D ensemble PTV”. In: *21st International Symposium on Applications of Laser and Imaging Techniques to Fluid Mechanics (LXLASER24)*. Lisbon, Portugal, 2024.

[38] P. A. Marques, A. Simonini, M. Delsipée, and M. A. Mendez. “Characterization of cryogenic sloshing via image velocimetry and free-surface detection with physics-constrained RBFs”. In: *21st International Symposium on Applications of Laser and Imaging Techniques to Fluid Mechanics (LXLASER24)*. Lisbon, Portugal, 2024.

[39] P. A. Marques, F. Monteiro, A. Simonini, L. Carbonnelle, and M. A. Mendez. “Data-driven thermodynamic modeling of microgravity sloshing: outcomes from the 83rd ESA parabolic flight campaign”. In: *28th European Low Gravity Research Association (ELGRA) Symposium*. Liverpool, United Kingdom, 2024.

[40] F. Monteiro, P. A. Marques, A. Simonini, L. Carbonnelle, and M. A. Mendez. “Experimental characterisation of sloshing-induced thermal mixing under gravity-dominated and microgravity conditions for space applications”. In: *28th European Low Gravity Research Association (ELGRA) Symposium*. Liverpool, United Kingdom, 2024.

[41] S. Randino, L. Schena, E. Gillyns, N. Coudou, and M. A. Mendez. "Real-time data assimilation for the digital twinning of wind farms". In: *1st European Fluid Dynamics Conference (EFDC1)*. Aachen, Germany, 2024.

[42] R. Antonissen and A. Mendez. "Reinforcement Twinning algorithms for dynamic propeller control". In: *1st European Fluid Dynamics Conference (EFDC1)*. Aachen, Germany, 2024.

[43] S. Ahizi, P. A. Marques, and M. A. Mendez. "A multi-environment formulation for data assimilation in cryogenic storage tanks". In: *European Fluid Dynamics Conference (EFDC1)*. Aachen, Germany, 2024.

[44] S. Ahizi, M. Fiore, L. Koloszar, and M. A. Mendez. "Data-driven, scale-adaptive turbulence-closure modeling". In: *European Fluid Dynamics Conference (EFDC1)*. Aachen, Germany, 2024.

[45] P. A. Marques, A. Simonini, M. Delsipée, and M. A. Mendez. "Experimental characterization of heat and mass transfer in cryogenic storage tanks". In: *9th World Congress on Momentum, Heat and Mass Transfer (MHMT 2024)*. London, United Kingdom, 2024. DOI: 10.11159/icmfht24.153.

[46] P. A. Marques, A. Simonini, M. Delsipée, and M. A. Mendez. "Model-based analysis of heat and mass transfer in cryogenic storage measurements". In: *3AF International Conference on Space Propulsion*. Submitted. Glasgow, United Kingdom, 2024.

[47] F. Monteiro, P. A. Marques, L. Carbonnelle, A. Simonini, and M. A. Mendez. "Non-isothermal sloshing for space applications: experimental characterization under reduced-gravity conditions". In: *3AF International Conference on Space Propulsion*. Submitted. Glasgow, United Kingdom, 2024.

[48] S. Ahizi, P. Marques, F. Monteiro, L. Gonzalez, F. Gambioli, H. Scheufler, R. Abarca, and M. A. Mendez. "Scaling laws of sloshing-induced thermodynamic fluctuations in a horizontal cylindrical tank". In: *International Ocean and Polar Engineering Conference (ISOPE)*. Rhodes, Greece, 2024.

[49] I. Tirelli, M. A. Mendez, A. Ianiro, and S. Discetti. "A combination of KNN-PTV and physics-constrained RBFs for super-resolution in image velocimetry". In: *15th International Symposium on Particle Image Velocimetry (ISPIV 2023)*. San Diego, USA, 2023.

[50] I. Tirelli, M. A. Mendez, A. Ianiro, and S. Discetti. "A combination of KNN-PTV and physics-constrained RBFs for super-resolution in image velocimetry". In: *20th International Symposium on Flow Visualization (ISFV20)*. Delft, Netherlands, 2023.

[51] M. Ratz, A. Simonini, and M. A. Mendez. "Physics-constrained and meshless data assimilation of three-dimensional particle tracking velocimetry". In: *20th International Symposium on Flow Visualization (ISFV20)*. Delft, Netherlands, 2023.

[52] D. Rigutto, J.-M. Buchlin, and M. A. Mendez. "Experimental analysis of dynamic menisci in dip coating". In: *20th International Symposium on Flow Visualization (ISFV20)*. Delft, Netherlands, 2023.

[53] D. Rigutto, J.-M. Buchlin, and M. A. Mendez. "Experimental analysis of dynamic menisci in dip coating". In: *15th European Coating Symposium (ECS 2023)*. Paris, France, 2023.

[54] D. Barreiro-Villaverde, A. Gosset, M. Lema, and M. A. Mendez. "High- and low-fidelity models to understand the physics of air-knife coating". In: *15th European Coating Symposium (ECS 2023)*. Paris, France, 2023.

[55] D. Barreiro-Villaverde, M. Lema, A. Gosset, and M. A. Mendez. "Implementation of a solver to simulate industrial galvanization". In: *V XoveTIC Conference (XoveTIC 2022)*. A Coruña, Spain, 2023.

[56] J. Van den Berghe, M. Delsipée, P. Planquart, Y. Bartosiewicz, and M. A. Mendez. "Calibration of a 1D ejector model from full-scale experiments". In: *8th Thermal and Fluids Engineering Conference (TFEC)*. 2023.

[57] J. Van den Berghe, B. R. Barros Dias, Y. Bartosiewicz, and M. A. Mendez. "Unsteady 1D gas dynamics in ejectors: a pipeline analogy". In: *22nd Computational Fluid Dynamics Conference (CFC)*. 2023.

[58] J. Van den Berghe, J. B. Vemula, Y. Bartosiewicz, and M. A. Mendez. "A machine-learning-based calibration of a 1D ejector model from CFD". In: *36th International Conference on Efficiency, Cost, Optimization, Simulation and Environmental Impact of Energy Systems (ECOS)*. 2023.

[59] S. Ahizi, P. Marques, and M. A. Mendez. "Modelling and scaling laws of cryogenic tank thermal response to sloshing". In: *36th International Conference on Efficiency, Cost, Optimization, Simulation and Environmental Impact of Energy Systems (ECOS)*. 2023.

[60] D. Fiorini, L. Carbonnelle, A. Simonini, J. Steelant, D. Seveno, and M. A. Mendez. "Characterization of a capillary-driven flow in microgravity by means of optical techniques". In: *11th International Conference on Multiphase Flow (ICMF 2023)*. 2023.

[61] D. Fiorini, A. Gosset, J. Steelant, D. Seveno, and M. A. Mendez. "On the capillary rise of low-surface-tension liquids in microgravity: numerical analysis and experiments". In: *10th European Conference for Aeronautics and Space Sciences (EUCASS)*. 2023.

[62] C. Esposito, M. A. Mendez, J. Steelant, and M. R. Vetrano. "Analyse modale multi-échelle pour l'identification des mécanismes clés caractérisant un écoulement cavitant". In: *17ème Congrès Francophone de Techniques Laser (CFTL)*. Leuven, Belgium, 2022.

[63] M. A. Mendez. "La décomposition modale pour l'analyse de visualisation d'écoulements". In: *17ème Congrès Francophone de Techniques Laser (CFTL)*. Leuven, Belgium, 2022.

[64] D. Meneghella, P. Marques, M. A. Mendez, and A. Simonini. "Comparaison d'amortissement libre de liquides soumis à ballottement par la POD". In: *17ème Congrès Francophone de Techniques Laser (CFTL)*. Leuven, Belgium, 2022.

[65] M. Ratz, S. Sachs, M. A. Mendez, and C. Cierpka. "Radial basis function regression of Lagrangian three-dimensional particle-tracking data". In: *20th International Symposium on Applications of Laser and Imaging Techniques to Fluid Mechanics*. Lisbon, Portugal, 2022.

[66] P. Sperotto, S. Pieraccini, and M. A. Mendez. "A RANS approach to the meshless computation of pressure fields from image velocimetry". In: *20th International Symposium on Applications of Laser and Imaging Techniques to Fluid Mechanics*. Lisbon, Portugal, 2022.

[67] F. Pino, B. Scheid, and M. A. Mendez. "Absolute/convective instability threshold in inverted falling film through linear stability analysis". In: *20th Conference of Numerical Analysis and Applied Mathematics (ICNAAM)*. Heraklion, Greece, 2022.

[68] C. Merhaben, B. Dias, M. A. Mendez, and A. Turchi. "Determination of trajectory-based ground-testing conditions for space-debris material by means of a data-driven approach". In: *FAR 2022 — 2nd International Conference on Flight Vehicles, Aerothermodynamics and Re-entry Missions and Engineering*. Heilbronn, Germany, 2022.

[69] D. Fiorini, M. A. Mendez, A. Simonini, D. Seveno, and J. Steelant. "Dynamic contact angle analysis using inverse methods". In: *10th International Conference on Inverse Problems in Engineering*. Malta, 2022.

[70] M. A. Mendez, J. Dominique, M. Fiore, F. Pino, P. Sperotto, and J. Van den Berghe. "Challenges and Opportunities for Machine Learning in Fluid Mechanics". In: *19th International Topical Meeting on Nuclear Reactor Thermal Hydraulics (NURETH)*. 2022.

[71] L. Schena, E. Gillyns, W. Munters, S. Buckingham, and M. A. Mendez. "Wind Turbine Control using Machine Learning techniques". In: *ECCOMAS 2022*. 2022.

[72] D. Barreiro-Villaverde, A. Gosset, M. A. Mendez, and M. Lema. "High-fidelity simulations of the wiping of low-viscosity liquid films". In: *14th European Fluid Mechanics Conference (EFMC14)*. 2022.

[73] D. Barreiro-Villaverde, A. Gosset, M. A. Mendez, and M. Lema. “Multiscale analysis of 3D VOF-LES simulations of the jet-wiping process”. In: *4th Iberian Meeting of OpenFOAM Users (Foam@Iberia 2022)*. 2022.

[74] D. Barreiro-Villaverde, A. Gosset, M. Lema, and M. A. Mendez. “Implementation of a solver to simulate industrial galvanization”. In: *5th XoveTIC Conference*. 2022.

[75] D. Barreiro-Villaverde, A. Gosset, M. A. Mendez, and M. Lema. “On the jet wiping of high-Kapitza liquid films”. In: *17th OpenFOAM Workshop (OFW17)*. 2022.

[76] D. Barreiro-Villaverde, A. Gosset, M. A. Mendez, and M. Lema. “High- and low-fidelity approaches to investigate instabilities in jet wiping”. In: *1st Spanish Fluid Mechanics Conference (SFMC22)*. 2022.

[77] D. Fiorini, L. Carbonnelle, A. Simonini, J. Steelant, D. Seveno, and M. A. Mendez. “Capillary rise in a divergent U-tube during parabolic flight”. In: *27th European Low Gravity Research Association (ELGRA) Biennial Symposium and General Assembly*. 2022.

[78] D. Fiorini, M. A. Mendez, A. Simonini, J. Steelant, and D. Seveno. “On the impact of contact-line acceleration in the spreading of perfectly wetting fluids”. In: *14th European Fluid Mechanics Conference (EFMC14)*. 2022.

[79] D. Fiorini, M. A. Mendez, A. Simonini, J. Steelant, and D. Seveno. “Dynamic wetting experiment with cryogenic nitrogen”. In: *Experimental Fluid Mechanics (EFM)*. 2022.

[80] F. Monteiro, P. Marques, A. Simonini, A. Silva, and M. A. Mendez. “Non-isothermal sloshing for space applications: from a ground-based experimental characterisation to microgravity conditions”. In: *27th European Low Gravity Research Association (ELGRA) Biennial Symposium and General Assembly*. 2022.

[81] M. Fiore, L. Koloszar, M. A. Mendez, M. Duponcheel, and Y. Bartosiewicz. “Artificial neural networks for the modeling of turbulent heat flux in forced-convection flows”. In: *19th International Topical Meeting on Nuclear Reactor Thermal Hydraulics (NURETH)*. 2022.

[82] M. Fiore, L. Koloszar, M. A. Mendez, M. Duponcheel, and Y. Bartosiewicz. “A physics-oriented data-driven approach to thermal-turbulence modeling”. In: *15th International Conference on Heat Transfer, Fluid Mechanics and Thermodynamics*. 2021.

[83] M. Ratz, D. Fiorini, A. Simonini, C. Cierpka, and M. A. Mendez. “Time-resolved PIV of the flow field underneath an accelerating meniscus”. In: *15th International Symposium on Particle Image Velocimetry (ISPIV 2021)*. 2021.

[84] F. Felis-Carrasco, D. Hess, B. B. Watz, and M. A. Mendez. “Investigation on computed pressures from PIV: how boundary definitions affect pressure accuracy”. In: *15th International Symposium on Particle Image Velocimetry (ISPIV 2021)*. 2021.

[85] T. Ivanova, F. Pino, and M. A. Mendez. “On the downstream evolution of three-dimensional waves in liquid films over moving substrate”. In: *European Coating Symposium (ECS 2021)*. Sint-Genesius-Rode, Belgium, 2021.

[86] D. Barreiro, A. Gosset, and M. A. Mendez. “On the undulation and the two-way coupled instability of the jet-wiping process”. In: *European Coating Symposium (ECS 2021)*. Sint-Genesius-Rode, Belgium, 2021.

[87] D. Fiorini, M. A. Mendez, A. Simonini, and D. Seveno. “Transient dynamics of contact angles”. In: *European Coating Symposium (ECS 2021)*. Sint-Genesius-Rode, Belgium, 2021.

[88] F. Pino, B. Scheid, and M. A. Mendez. “Modelling of magnetic wiping via integral models”. In: *European Coating Symposium (ECS 2021)*. Sint-Genesius-Rode, Belgium, 2021.

[89] M. Ratz, D. Fiorini, A. Simonini, C. Cierpka, and M. A. Mendez. “Dynamics of a quasi-capillary channel flow in accelerating conditions”. In: *European Coating Symposium (ECS 2021)*. Sint-Genesius-Rode, Belgium, 2021.

[90] E. Anfuso, S. Demange, A. Fagnani, F. T. Herrador, M. A. Mendez, and O. Chazot. "Multiscale modal analysis of a plasma jet: coherent structures and their observability". In: *AIAA Aviation Forum*. 2021.

[91] D. Fiorini, M. A. Mendez, A. Simonini, D. Seveno, and J. Steelant. "Dynamic contact angle analysis using inverse methods". In: *10th International Conference on Inverse Problems in Engineering*. Chieti, Italy, 2021.

[92] D. Barreiro-Villaverde, A. Gosset, M. A. Mendez, and J.-M. Buchlin. "Validation of high-fidelity CFD simulations of the jet-wiping process". In: *European Coating Symposium (ECS 2019)*. Heidelberg, Germany, 2019.

[93] M. A. Mendez, A. Gosset, B. Scheid, and J.-M. Buchlin. "Integral boundary-layer simulations of the jet-wiping process". In: *European Coating Symposium (ECS 2019)*. Heidelberg, Germany, 2019.

[94] M. A. Mendez and J.-M. Buchlin. "Multiscale proper orthogonal decomposition (mPOD) of TR-PIV data: a case study on transient flows". In: *13th International Symposium on Particle Image Velocimetry*. Munich, Germany, 2018.

[95] M. A. Mendez and J.-M. Buchlin. "Analyse du procédé d'essorage au moyen de la vélocimétrie par image de particules et la technique d'absorption de lumière". In: *16e Congrès Francophone de Techniques Laser (CFTL)*. Dourdan, France, 2018.

[96] M. A. Mendez, A. Enache, A. Gosset, and J.-M. Buchlin. "Experimental characterization of the jet-wiping process". In: *Experimental Fluid Mechanics (EFM)*. Mikulov, Czech Republic, 2017.

[97] M. A. Mendez, M. T. Scelzo, A. Enache, and J.-M. Buchlin. "Fluidic vectoring of a planar incompressible jet flow". In: *Experimental Fluid Mechanics (EFM)*. Mikulov, Czech Republic, 2017.

[98] A. Gosset, M. A. Mendez, and J.-M. Buchlin. "Experimental characterization of the jet-wiping instability". In: *European Coating Symposium (ECS 2017)*. Fribourg, Switzerland, 2017.

[99] M. A. Mendez, A. Gosset, K. Myrillas, and J.-M. Buchlin. "Numerical modal analysis of the jet-wiping instability". In: *European Coating Symposium (ECS 2017)*. Fribourg, Switzerland, 2017.

[100] M. A. Mendez, M. Balabane, and J.-M. Buchlin. "Multiscale modal analysis of experimental and numerical data (keynote, invited)". In: *Experiments in Fluid Mechanics*. Warsaw, Poland, 2017.

[101] M. A. Mendez, M. Balabane, and J.-M. Buchlin. "Multi-scale proper orthogonal decomposition (mPOD)". In: *International Conference of Numerical Analysis and Applied Mathematics (ICNAAM)*. Thessaloniki, Greece, 2017.

[102] M. A. Mendez, B. Scheid, and J.-M. Buchlin. "Low-Kapitza falling liquid films". In: *13th International Conference on Gas–Liquid and Gas–Liquid–Solid Reactor Engineering (GLS)*. Brussels, Belgium, 2017.

[103] M. A. Mendez, M. Raiola, A. Masullo, S. Discetti, A. Ianiro, R. Theunissen, and J.-M. Buchlin. "Prétraitement d'images PIV par décomposition aux valeurs propres". In: *15e Congrès Francophone de Techniques Laser (CFTL)*. Toulouse, France, 2016.

[104] M. A. Mendez and J.-M. Buchlin. "Oscillation of a planar gas jet impinging on a deformable wall". In: *Symposium of VKI PhD Research*. Brussels, Belgium, 2016.

[105] M. A. Mendez and J.-M. Buchlin. "Experimental characterization of 2D travelling waves in low-Kapitza liquid film down a vertical wall". In: *24th International Congress of Theoretical and Applied Mechanics (ICTAM 2016)*. Montreal, Canada, 2016.

[106] M. A. Mendez, L. Nemeth, and J.-M. Buchlin. "Measurement of liquid-film thickness via light absorption and laser tomography". In: *Experimental Fluid Mechanics (EFM 2015)*. Prague, Czech Republic, 2015.

[107] M. A. Mendez, K. Myrillas, A. Gosset, and J.-M. Buchlin. “A research methodology to study jet-wiping processes”. In: *European Coating Symposium (ECS 2015)*. Eindhoven, Netherlands, 2015.

[108] M. A. Mendez and J.-M. Buchlin. “Quantitative flow visualization of confinement-driven instabilities of an impinging slot jet”. In: *11th International Symposium on Particle Image Velocimetry (PIV 2015)*. Santa Barbara, USA, 2015.

[109] M. A. Mendez and J.-M. Buchlin. “Experimental characterization of straw jets via image-processing techniques”. In: *10th Pacific Symposium on Flow Visualization and Image Processing (PSFVIP 2015)*. Naples, Italy, 2015.

[110] M. A. Mendez and J.-M. Buchlin. “Dynamics of gas-jet impingement on vertical falling liquid films”. In: *Symposium of VKI PhD Research*. Brussels, Belgium, 2015.